



LESSON 9.7 DILATIONS

LESSON INTRODUCTION

MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.

LEARNING TARGETS

- I can describe the effect on an image given the scale factor.
- I can relate dilations to proportions.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

ESSENTIAL QUESTIONS

- How does the center of dilation impact the image?
- How does the scale factor impact the image?
- How are dilations related to proportions?

LESSON NARRATIVE

Students explore artwork by M.C. Escher to consider how the artwork shows transformations. Students explore properties of dilations by first examining the effect the center of dilation has on the location of the image. Students continue to explore properties of dilations by examining the effect the scale factor has on an image and how the distance the figures are from the center are related to proportions.

COMPONENT SUMMARY

9.7.1 WARM-UP (~5 MIN)

Place points scale distances on a ray

9.7.3 EXPLORATION (25 MIN)

Explore properties of dilations

9.7.5 WRAP-UP (~5 MIN)

Reflect on mastery of learning targets

9.7.2 COLLABORATION (10 MIN)

Identify rigid and non-rigid transformations in tessellations

9.7.4 BRING IT TOGETHER (5 MIN)

Synthesize the properties of dilations

RESOURCES

GLOSSARY

Key Terms:

- Proportional
- Scale factor
- Dilation

Additional Terms: N/A

MATERIALS

- Rulers
- Patty Paper
- Calculators
- (Optional) Perspective Drawings

ADDITIONAL PREPARATION

- Consider creating a Frayer Model with definitions, examples, non-examples, and visuals of dilations.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think figures are dilations if they are bigger or smaller and may not realize the preimage and image must be proportional for dilations.
- Students may think that dilations are rigid transformations.

THINGS TO CONSIDER

- Consider using tessellations to illustrate how dilations are not rigid transformations.
- Consider connecting dilations to lines of perspective in art.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 9.7.3 Exploration students work with several sets of dilations. Ask students what they notice and wonder about each dilation. Encourage students to conjecture about the relationship between the preimages and images.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

While pairs are working during 9.7.3 Exploration, circulate and take note of which students you will call on to share during the debrief. During the debrief, create or reference a Frayer Model with the vocabulary, phrases, and visuals students used.

DIFFERENTIATE INSTRUCTION

The discussions during 9.7.3 Exploration provide an auditory entry-point. 9.7.2 Collaboration provides a visual entry-point as students see examples of rigid and non-rigid transformations tessellations with. Students may also benefit from physically touching scale models, such as a model car, Russian doll, or measuring spoons.



9.7.1 WARM-UP: POINTS ON A RAY

TEACHER MOVES

Students can measure the distance from A to B using a ruler, mark the distance on a paper, or trace the line segment on tracing paper. Have students share the different strategies they used. Discuss how the lengths of line segments (AB) , (AC) , and (AD) compare. Note any key words that students use in their descriptions.

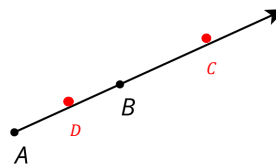
DIFFERENTIATE INSTRUCTION

Consider clarifying the vocabulary term “scale factor” as students provide strategies and descriptions.



9.7.1 Warm-Up: Points on a Ray

1. A ray is shown with points A and B labeled.



- a. Find a point on the ray whose distance from A is twice the distance from B to A . Label the point C .
- b. Find a point on the ray whose distance from A is half the distance from B to A . Label the point D .
- c. Use the emoji scale to rate your confidence in your placement of points C and D on the ray.

Answers will vary.

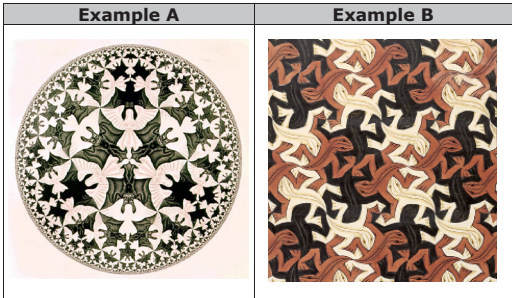




9.7.2 Collaboration: Artwork of M.C. Escher

M.C. Escher (1898-1972) was a Dutch artist whose work was inspired by mathematics. Though initially trained as an architect, he became one of the most famous and influential graphic artists in the world.

Below are two examples of Escher’s work.



1. With your partner, discuss how the two example pieces shown are similar and different. Then, record at least two in each column.

Similarities	Differences
Answers will vary. Both pieces have images that are repeated. In both pieces, the repeated images are rotated.	Answers will vary. Example A is black and white, and Example B is in color. In Example B, all the geckos are the same size, but in Example A, the bats and angels are different sizes.

2. Consider how the figures used to make the artwork were transformed.
- a. Which of the art pieces could be created using only rigid transformations of the shapes used?
Example B

- b. Explain how you know the other example could not be created from rigid transformations.
- Answers will vary. I know Example A could not be created from rigid transformations because the objects are not all the same size, so they are not congruent.*

9.7.2 COLLABORATION: ARTWORK OF M.C. ESCHER

TEACHER MOVES

Consider giving students a chance to turn and talk with a partner about their initial reactions to Escher’s tessellations.

ENRICHMENT

- What do you notice about the artwork? How do you think the artist may have created them?
- What transformations did he use?

DIFFERENTIATE INSTRUCTION

Provide students the opportunity to view enlarged versions of Escher’s tessellations by projecting them in the classroom.

QUESTIONING

Students can create their own tessellations. Ask students to note details about the transformations they use while creating their tessellations.

9.7.3 EXPLORATION: PROPERTIES OF DILATIONS

TEACHER MOVES

Notice and Wonder

As students work with several sets of dilations ask them what they notice and wonder about each dilation. Encourage students to conjecture about the relationship between the preimages and images.

TEACHER MOVES

Consider having students align on a set of vertices with the center of dilation using a ruler. This may help students understand the effect the center of dilation has with the preimage and image. Consider asking these questions:

- How are the vertices of the figures related to the center of dilation?

- What do you notice about the center of dilation in Transformation D?

TEACHER MOVES

Four transformations are shown with question 1. If students would benefit from seeing additional examples to observe what happens when the center of dilation is moved but the scale factor remains the same, consider using the optional GeoGebra link for question 1 found in the digital Teacher Edition to demonstrate additional transformations with these constraints.

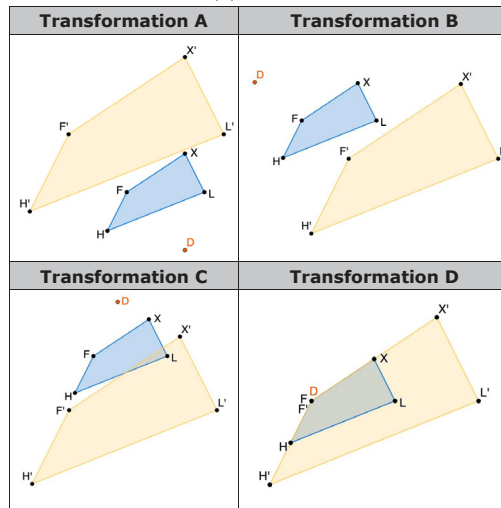


9.7.3 Exploration: Properties of Dilations

Another type of transformation is called a **dilation**. Dilations are not rigid transformations because the size of the image can change from the preimage.

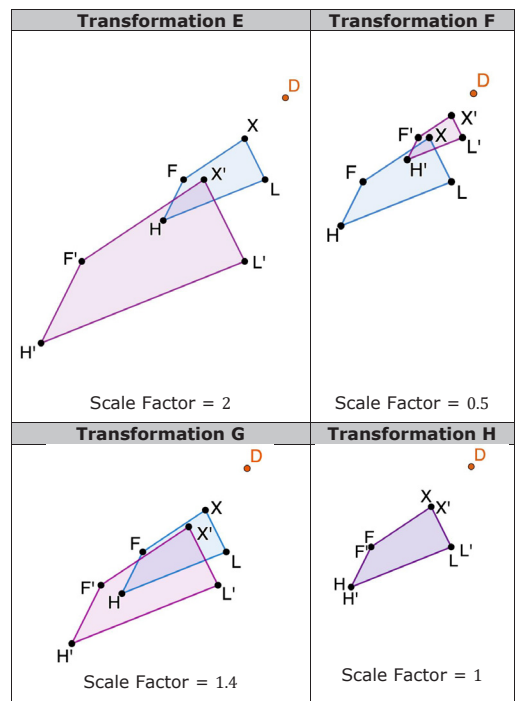
To dilate a figure, both a center and a scale factor must be given. Work with your partner to complete the following as you explore the properties of dilations.

- Each of the transformations shown is a dilation of the preimage, quadrilateral $FXLH$, using the same scale factor, but a different center of dilation. The center of dilation is indicated by point D .



- Discuss with your partner what you notice about each dilation.
- Describe what happens to the image, quadrilateral $F'X'L'H'$, as the center of dilation, Point D , changes.
Answers will vary. As the center of dilation, Point D , changes, the image, quadrilateral $F'X'L'H'$ stays the same size, shape, and orientation, but changes location.
- In which transformation is the center of dilation a point on the preimage?
Transformation D
- Describe how this dilation appears to be different from the others.
Answers will vary. Transformation D looks different from the others because the image is completely on top of the preimage.

2. Each of the transformations shown is a dilation of the preimage, quadrilateral $FXLH$, using the same center of dilation, but a different scale factor. The center of dilation is indicated by point D , and the scale factor is given in the table.

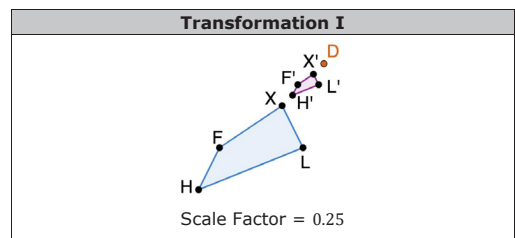


9.7.3 EXPLORATION: PROPERTIES OF DILATIONS (CONTINUED)

TEACHER MOVES

Consider having students align one set of vertices with the center of dilation using a ruler. Consider asking these questions:

- How is the distance from the center of dilation for the preimage and image related to the scale factor?
- What do you notice about the distance from the center of dilation for the preimage and image in Transformation H?



- a. Discuss with your partner what happens to the image, quadrilateral $FXLH$, as the scale factor changes.
- b. Complete the table.

Value of the Scale Factor	Between 0 and 1	Equal to 1	Greater than 1
How does the image compare to the preimage?	The image is smaller than the preimage and closer to the center of dilation.	The image and the preimage are exactly the same.	The image is larger than the preimage and farther from the center of dilation.

TEACHER MOVES

While pairs are working, circulate and take note of which students you will call on to share during the debrief. During the debrief, create or reference a Frayer Model with the vocabulary, phrases, and visuals students used.

DIFFERENTIATE INSTRUCTION

Consider projecting the dilations can see them more clearly. The projected dilations will be easier to reference during the debrief discussion.

9.7.3 EXPLORATION: PROPERTIES OF DILATIONS (CONTINUED)

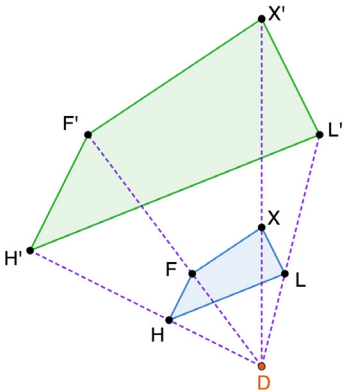
TEACHER MOVES

Consider asking any students who take art how the lines from the center of dilation through the vertices are related to perspective drawing. Also consider showing artwork that uses perspective drawing. Confer with an art teacher about how perspective drawings create the appearance of size.

QUESTIONING

To connect dilations to art, consider providing students with perspective grid to create a drawing. The drawing can use simple shapes or be more complicated.

3. The transformation shows a dilation of quadrilateral $FXLH$ with the center of dilation at point D and a scale factor of 2.5.



- a. The distance, in units, between the center of dilation, D , and each of the vertices of the preimage and image are given in the table. Work with your partner to complete the table.

Vertex	Distance from Point D	Ratio $\left(\frac{\text{image}}{\text{preimage}}\right)$ of the Distances
H'	11.25	$\frac{11.25}{4.5} = 2.5$
H	4.5	
F'	10.25	$\frac{10.25}{4.1} = 2.5$
F	4.1	
X'	12.5	$\frac{12.5}{5} = 2.5$
X	5	
L'	15	$\frac{15}{6} = 2.5$
L	6	

- b. Find the value of each ratio in the table using your calculator. What do you notice?
All the ratios are equal to 2.5.

- c. Explain how the ratios of the distances compare to the scale factor of the dilation.
The ratios of the distances between the center of dilation, D , and each of the vertices of the preimage and image is equal to the scale factor of dilation.



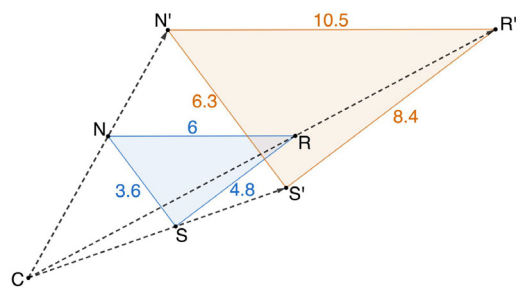
9.7.4 Bring it Together: Properties of Dilations

Every dilation has a **center of dilation** and a **scale factor**.



In a dilation, the distance of the image to the center of dilation is **proportional** to the distance of the preimage to the center of dilation. The constant of proportionality is equal to the **scale factor**.

- Triangle NRS was dilated using a scale factor of 1.75 centered at point C . The side lengths, in units, of each triangle are shown.



- Trace $\triangle NRS$ on patty paper, labeling each vertex on the patty paper.
- Overlay the patty paper on $\triangle N'R'S'$ at each vertex to compare the size of each corresponding angle. What do you notice?
Corresponding angles have the same measure.

9.7.4 BRING IT TOGETHER: PROPERTIES OF DILATIONS

TEACHER MOVES

Prompt prior knowledge by asking students where scale factor is used. Students first learn about scale factor in 7th grade. Also engage students in what they know and understand about the word proportional and what makes two things proportional.

DIFFERENTIATE INSTRUCTION

Tracing the triangles on patty paper and overlaying the angles of the image on the angles of the preimage provides a kinesthetic entry-point.

9.7.4 BRING IT TOGETHER: PROPERTIES OF DILATIONS (CONTINUED)

ENRICHMENT

- How does the scale factor impact whether the image will be increased or decreased in size from the preimage?
- When will the image be the same size as the preimage?

TEACHER MOVES

Students explored similar figures in Grade 8 Unit 6. Facilitate student discussion about the connections between similar figures and dilations.

- c. Complete the table using the side lengths of the image and preimage.

Side	Length	Side	Length	Ratio of the Side Lengths, $\frac{\text{Image}}{\text{Preimage}}$
NR	6	$N'R'$	10.5	$\frac{10.5}{6} = 1.75$
RS	4.8	$R'S'$	8.4	$\frac{8.4}{4.8} = 1.75$
SN	3.6	$S'N'$	6.3	$\frac{6.3}{3.6} = 1.75$

- d. What do you notice about the side lengths of $\triangle NRS$ and $\triangle N'R'S'$?

Side lengths are proportional, as the ratios between them are equivalent.

- e. Complete the statements using words from the bank.

congruent proportional similar

After a dilation, the corresponding side lengths of the preimage and image are proportional and the corresponding angles are congruent. Therefore, the preimage and image are similar figures.



A **dilation** is a proportional increase or decrease in size in all directions.



9.7.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information in the Exploration and Bring it Together that I knew were ...	Something I wondered during the Exploration or Bring it Together was ...	Something I learned during the Exploration or Bring it Together was ...
<i>Answers will vary.</i>	<i>Answers will vary.</i>	<i>Answers will vary.</i>

TEACHER MOVES

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

DIFFERENTIATE INSTRUCTION

Allow students to respond in writing, verbally, or by creating visual representations.